Lab Report

BA-MECH-24-4C

Control Engineering II

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Chapter 1

Introduction

1.1 General Description

This report documents the modelling, identification and position control of a DC motor that drives a pointer via a gearbox with belt drive (Box 1 of the laboratory test benches). The controlled variable is the angular position of the pointer, measured on the output shaft by an optical quadrature encoder. The motor is supplied by a PWM-controlled motor driver, and all signals are processed by a data acquisition card with MATLAB/Simulink and Simulink Desktop Real-Time [1].

1.2 Motivation

DC motors are widely used electro-mechanical actuators, for example in automotive applications, medical devices and industrial automation [1]. An exact positioning of the driven load requires a closed-loop controller, whose design relies on a sufficiently accurate model of the plant. Since not all parameters of a real drive train are known, the model is derived both analytically and from measured data.

1.3 Learning Goal

The aim of the laboratory is to identify an electro-mechanical system from its measured step response and to design a suitable controller [1]. The step response is approximated by a second-order lag element, a PID controller is designed with the Ziegler–Nichols method, and the closed loop is simulated, implemented on the real test bench and evaluated.

Chapter 2

Procedure and Method

2.1 Experimental Setup

The equipment used for the experiment is listed in Table 2.1.

Table 2.1: List of equipment used for the experiment [1]

Component	Description	Type
Test bench	gearbox with belt drive, Box 1, $N = 97.2$	–
DC motor	brushed DC motor, nominal voltage $12\,\mathrm{V}$	Faulhaber 3257G012CR [2]
Motor driver	PWM-controlled motor driver	Cytron MD13S [3]
Encoder	quadrature encoder, 500 counts per revolution	Broadcom HEDS-5500 A11 [4]
DAQ card	data acquisition card	NI PCIe-6321 [5]
I/O module	BNC connector block	NI BNC-2090a
Software	identification, simulation, real-time control	MATLAB/Simulink

2.2 Analytical Modelling of the DC Motor

The unloaded DC motor is described by a white-box model based on Figure 2.1. In this report, the motor quantities are written without the index "mot", U_{in} is denoted by u, U_{emf} by u_{emf}, and the rotation n_{out} is described by the angular velocity ω.

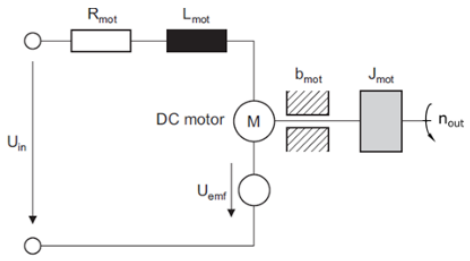


Figure 2.1: Schematic representation of the electro-mechanical system of the DC motor [1]

Kirchhoff's voltage law for the armature circuit with the back electromotive force $u_{\mathrm{emf}} = k_{\mathrm{E}}\,\omega$ yields Equation (2.1), the balance of torques at the rotor yields Equation (2.2).

$$u(t) = R\,i(t) + L\,\frac{\mathrm{d}i(t)}{\mathrm{d}t} + k_{\mathrm{E}}\,\omega(t) \tag{2.1}$$

$$J\,\frac{\mathrm{d}\omega(t)}{\mathrm{d}t} = k_{\mathrm{M}}\,i(t) - b\,\omega(t) \tag{2.2}$$

u	armature voltage, in V	J	rotor inertia, in $\mathrm{kg\,m^2}$
i	armature current, in A	b	viscous friction, in $\mathrm{N\,m\,s\,rad^{-1}}$
ω	angular velocity, in $\mathrm{rad\,s^{-1}}$	k_{M}	torque constant, in $\mathrm{N\,m\,A^{-1}}$
R	armature resistance, in Ω	k_{E}	back-EMF constant, in $\mathrm{V\,s\,rad^{-1}}$
L	armature inductance, in H	t	time, in s

With $\omega = \mathrm{d}\theta/\mathrm{d}t$ for the angular position θ, the Laplace transformation (capital letters, Laplace variable s) yields the transfer function from the voltage to the angular velocity, which is a second-order lag element (PT2), and to the angular position, which additionally contains an integrator. The derivation, both transfer functions and the state-space model are given in Appendix A.

2.3 System Identification

The step response is recorded for an input step of $u_0 = 12\,\mathrm{V}$ at $t = 0\,\mathrm{s}$ with a sample time of $0.1\,\mathrm{ms}$ [1]. The encoder provides the rotational speed n of the output shaft in $\mathrm{min^{-1}}$ (resolution $1\,\mathrm{min^{-1}}$), which is converted into $\omega = 2\pi n/60\,\mathrm{s\,min^{-1}}$; ω thus refers to the output shaft, which is slower than the motor shaft by the factor N. The plant is approximated by an overdamped PT2

element according to Equation (2.3) with the static gain K in $\mathrm{rad\,s^{-1}\,V^{-1}}$ and the time constants τ_1, τ_2 in s.

$$G_{\mathrm{PT2}}(s) = \frac{\Omega(s)}{U(s)} = \frac{K}{(\tau_1 s + 1)(\tau_2 s + 1)} \tag{2.3}$$

K, τ_1 and τ_2 are determined by a least-squares fit of the step response

$$\omega(t) = u_0 K \left[1 - \frac{\tau_1 \,\mathrm{e}^{-t/\tau_1} - \tau_2 \,\mathrm{e}^{-t/\tau_2}}{\tau_1 - \tau_2}\right]$$

to the measurement with the MATLAB function `lsqcurvefit`. Integrating the angular velocity gives the plant for the position control in Equation (2.4) [1].

$$G_{\mathrm{p}}(s) = \frac{\Theta(s)}{U(s)} = \frac{1}{s}\, G_{\mathrm{PT2}}(s) = \frac{K}{s\,(\tau_1 s + 1)(\tau_2 s + 1)} \tag{2.4}$$

2.4 Controller Design and Implementation

A PID controller according to Equation (2.5) with the proportional gain K_{P} in $\mathrm{V\,rad^{-1}}$, the integral and derivative times T_{I}, T_{D} in s, $K_{\mathrm{I}} = K_{\mathrm{P}}/T_{\mathrm{I}}$ and $K_{\mathrm{D}} = K_{\mathrm{P}}\,T_{\mathrm{D}}$ is designed with the closed-loop method of Ziegler and Nichols [6].

$$G_{\mathrm{c}}(s) = K_{\mathrm{P}} \left(1 + \frac{1}{T_{\mathrm{I}}\, s} + T_{\mathrm{D}}\, s\right) = K_{\mathrm{P}} + \frac{K_{\mathrm{I}}}{s} + K_{\mathrm{D}}\, s \tag{2.5}$$

The critical gain K_{cr}, at which the closed loop of $G_{\mathrm{p}}(s)$ with a pure P controller reaches the stability limit, is determined from the root locus with the MATLAB function `rlocus`, and the period P_{cr} of the sustained oscillation from the simulated output signal [1]. The tuning rules $K_{\mathrm{P}} = 0.6\,K_{\mathrm{cr}}$, $T_{\mathrm{I}} = 0.5\,P_{\mathrm{cr}}$ and $T_{\mathrm{D}} = 0.125\,P_{\mathrm{cr}}$ then give the controller parameters [6]. The closed loop is simulated in Simulink, with the reference converted from degrees into radians before the controller and the output converted back [1].

The controller is implemented in Simulink Desktop Real-Time with a sample time of $0.1\,\mathrm{ms}$, in the parallel form of Equation (2.5) with a first-order derivative filter (coefficient $100\,\mathrm{s^{-1}}$). The control value is limited to $\pm 12\,\mathrm{V}$, and integrator windup is prevented by clamping. For each run, a reference step of $45°$ is recorded for $1\,\mathrm{s}$, while K_{P}, K_{I} and K_{D} are varied one after another. In addition, a disturbance applied to the pointer and the tracking of a sinusoidal reference are recorded.

Chapter 3

Results

3.1 Identified Transfer Function

The least-squares fit yields $K = 0.433\,\mathrm{rad\,s^{-1}\,V^{-1}}$, $\tau_1 = 7.6\,\mathrm{ms}$ and $\tau_2 = 0.103\,\mathrm{s}$ with a coefficient of determination of 0.996 and a root-mean-square deviation of $0.064\,\mathrm{rad\,s^{-1}}$. Measurement and model are compared in Figure 3.1.

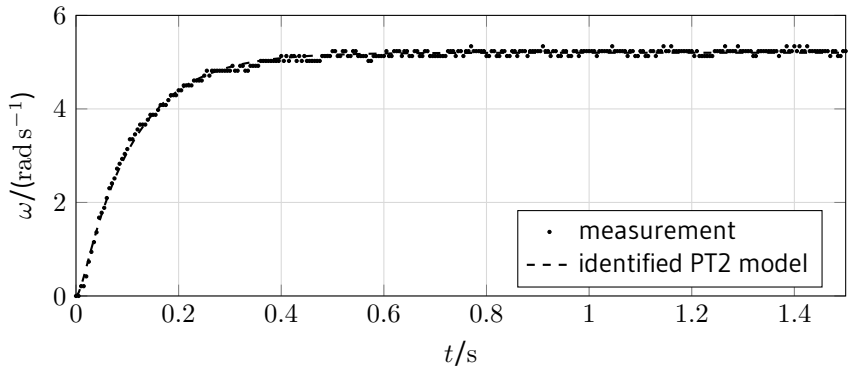


Figure 3.1: Measured angular velocity of the output shaft (dots, every $5\,\mathrm{ms}$) and step response of the identified PT2 model (dashed) for an input step of $12\,\mathrm{V}$

As a plausibility check, the static gain is calculated from the mean rotational speed $n_\infty = 49.8\,\mathrm{min^{-1}}$ for $t \geq 0.8\,\mathrm{s}$:

$$\omega_\infty = \frac{2\pi \cdot 49.8\,\mathrm{min^{-1}}}{60\,\mathrm{s\,min^{-1}}} = 5.22\,\mathrm{rad\,s^{-1}}, \qquad K = \frac{\omega_\infty}{u_0} = \frac{5.22\,\mathrm{rad\,s^{-1}}}{12\,\mathrm{V}} = 0.435\,\mathrm{rad\,s^{-1}\,V^{-1}}.$$

This deviates by less than $1\,\%$ from the fitted gain. With Equation (2.4), the plant for the position

control is

$$G_{\mathrm{p}}(s) = \frac{0.433\,\mathrm{rad\,s^{-1}\,V^{-1}}}{s\,(0.0076\,\mathrm{s}\cdot s + 1)(0.103\,\mathrm{s}\cdot s + 1)}.$$

3.2 Controller Parameters and Simulation

The root locus of $G_{\mathrm{p}}(s)$ yields a critical gain of $K_{\mathrm{cr}} = 325\,\mathrm{V\,rad^{-1}}$ and a period of $P_{\mathrm{cr}} = 0.176\,\mathrm{s}$. The tuning rules give $K_{\mathrm{P}} = 0.6 \cdot 325\,\mathrm{V\,rad^{-1}} = 195\,\mathrm{V\,rad^{-1}}$, $T_{\mathrm{I}} = 0.5 \cdot 0.176\,\mathrm{s} = 0.0879\,\mathrm{s}$ and $T_{\mathrm{D}} = 0.125 \cdot 0.176\,\mathrm{s} = 0.0220\,\mathrm{s}$, and thus

$$K_{\mathrm{I}} = \frac{K_{\mathrm{P}}}{T_{\mathrm{I}}} = \frac{195\,\mathrm{V\,rad^{-1}}}{0.0879\,\mathrm{s}} = 2.22 \times 10^{3}\,\mathrm{V\,rad^{-1}\,s^{-1}}$$

and $K_{\mathrm{D}} = K_{\mathrm{P}}\,T_{\mathrm{D}} = 195\,\mathrm{V\,rad^{-1}} \cdot 0.0220\,\mathrm{s} = 4.28\,\mathrm{V\,s\,rad^{-1}}$.

3.3 Real-Time Control

Table 3.1 summarises the controller gains of all runs and the characteristics of the step responses: the rise time t_{r} between $10\,\%$ and $90\,\%$ of the reference, the settling time t_{s} for a tolerance band of $\pm 2\,\%$, and the control error e (reference minus actual position) at the end of the recording. The smallest position increment of the encoder is $0.18°$.

Table 3.1: Controller gains and characteristics of the measured step responses for a reference step of $45°$ (in runs 3 and 4, the position is still changing at the end of the recording after $1\,\mathrm{s}$)

Run	K_{P}/(V rad^{-1})	K_{I}/(V rad^{-1} s^{-1})	K_{D}/(V s rad^{-1})	Overshoot/%	t_{r}/s	t_{s}/s	e/°
2	195	2219.7	4.28	3.2	0.173	0.408	0.0
1	195	22.2	4.28	0.0	0.175	0.291	0.0
3	1	22.2	4.28	–	0.688	–	−6.3
4	20	22.2	4.28	–	0.472	–	−5.4
5	195	22.2	8.00	0.0	0.195	0.347	0.0
6	195	22.2	1.00	4.8	0.173	0.311	0.0
7	300	22.2	4.28	0.4	0.173	0.251	0.0
8	300	22.2	6.00	0.0	0.172	0.272	0.0

With the calculated Ziegler–Nichols parameters (run 2), the position overshoots by $3.2\,\%$ and settles

after $0.41\,\mathrm{s}$. Reducing K_I by a factor of 100 (run 1) removes the overshoot and shortens the settling time to $0.29\,\mathrm{s}$; run 1 is the reference for the further variations. Figure 3.2 compares both runs with a strongly reduced K_P (run 3).

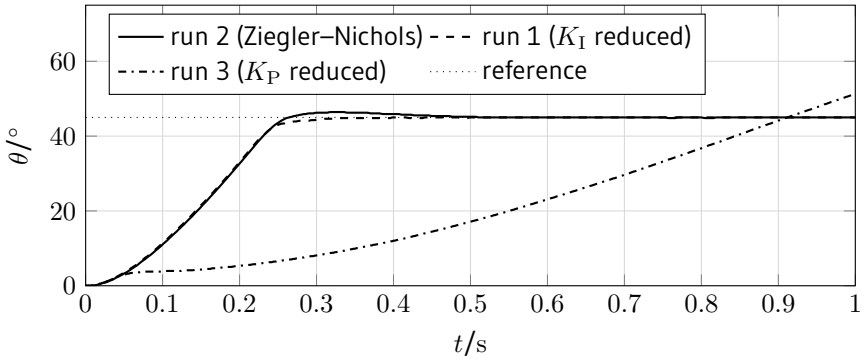


Figure 3.2: Measured angular position of the pointer for a reference step of $45°$ with the calculated Ziegler–Nichols parameters (solid), reduced integral gain (dashed) and reduced proportional gain (dash-dotted); gains see Table 3.1

With $K_\mathrm{P} = 1\,\mathrm{V\,rad^{-1}}$ and $20\,\mathrm{V\,rad^{-1}}$ (runs 3 and 4), the position does not settle within the recording and still exceeds the reference by $6.3°$ and $5.4°$ at $t = 1\,\mathrm{s}$. A larger K_D (run 5) prevents the overshoot but slows the response, a smaller K_D (run 6) causes $4.8\,\%$ overshoot. The shortest settling time of $0.25\,\mathrm{s}$ is reached with $K_\mathrm{P} = 300\,\mathrm{V\,rad^{-1}}$ (run 7) at an overshoot of one encoder increment, which vanishes with a larger K_D (run 8). In all runs, the control value saturates at $12\,\mathrm{V}$ directly after the step. In the disturbance test with the gains of run 1, the pointer is deflected by up to $2.2°$ and returns to within $0.5°$ of the reference after approximately $1.7\,\mathrm{s}$. For the tracking of a sinusoidal reference (Table B.1 in Appendix B), increasing K_P from $195\,\mathrm{V\,rad^{-1}}$ to $1200\,\mathrm{V\,rad^{-1}}$ reduces the RMS control error from $0.59°$ to $0.12°$, while the maximum control value rises from $3.5\,\mathrm{V}$ to $10.6\,\mathrm{V}$ and becomes increasingly noisy.

Chapter 4

Discussion and Interpretation

4.1 Comparison of Measurement, Model and Simulation

4.2 Controller Performance

4.3 Conclusion

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List of Symbols

Symbol	Description	Unit
$\mathbf{A}$	system matrix of the state-space model	–
b	viscous friction coefficient	$\mathrm{N\,m\,s\,rad^{-1}}$
$\mathbf{b}$	input vector of the state-space model	–
$\mathbf{c}$	output vector of the state-space model	–
d	feedthrough of the state-space model	–
e	control error (reference minus actual position)	$^\circ$
G	transfer function	–
i	armature current	A
J	moment of inertia of the rotor	$\mathrm{kg\,m^2}$
k	motor constant (k_M: torque constant, k_E: back-EMF constant)	$\mathrm{N\,m\,A^{-1}}$, $\mathrm{V\,s\,rad^{-1}}$
K	gain (K: static gain of the plant; K_P, K_I, K_D: controller gains; K_cr: critical gain)	–
L	armature inductance	H
n	rotational speed of the output shaft	$\mathrm{min^{-1}}$
N	transmission ratio of the gearbox	1
P	period of oscillation	s
R	armature resistance	Ω
s	Laplace variable	$\mathrm{s^{-1}}$
t	time	s
T	controller time constant (T_I: integral, T_D: derivative)	s
u	armature voltage	V
$\mathbf{x}$	state vector	–
θ	angular position	rad
τ	time constant of the PT2 model	s
ω	angular velocity	$\mathrm{rad\,s^{-1}}$

Chapter A

Derivation of the Analytical Model

Laplace transformation of Equations (2.1) and (2.2) with zero initial conditions and $\Omega(s) = s\,\Theta(s)$ gives

$$U(s) = (R + Ls)\,I(s) + k_\mathrm{E}\,s\,\Theta(s), \qquad Js^2\,\Theta(s) = k_\mathrm{M}\,I(s) - b\,s\,\Theta(s).$$

Solving the mechanical equation for the current,

$$I(s) = \frac{(Js^2 + bs)\,\Theta(s)}{k_\mathrm{M}},$$

and substituting it into the electrical equation yields

$$\begin{aligned} U(s) &= \left[\frac{(R + Ls)(Js^2 + bs)}{k_\mathrm{M}} + k_\mathrm{E}\,s\right]\Theta(s) \\ &= \frac{s\left[LJs^2 + (RJ + Lb)\,s + (Rb + k_\mathrm{E}k_\mathrm{M})\right]}{k_\mathrm{M}}\,\Theta(s), \end{aligned}$$

from which the transfer function from the armature voltage to the angular position in Equation (A.1) follows. It is a third-order system with an integrator, since the angular position is the integral of the angular velocity.

$$G_{\mathrm{p1}}(s) = \frac{\Theta(s)}{U(s)} = \frac{k_\mathrm{M}}{s\left[LJs^2 + (RJ + Lb)\,s + (Rb + k_\mathrm{E}k_\mathrm{M})\right]} \tag{A.1}$$

With $\Omega(s) = s\,\Theta(s)$, the transfer function from the armature voltage to the angular velocity in Equation (A.2) is a second-order lag element (PT2), which justifies the PT2 approach in Section 2.3.

$$G_{\mathrm{p2}}(s) = \frac{\Omega(s)}{U(s)} = s\,G_{\mathrm{p1}}(s) = \frac{k_\mathrm{M}}{LJs^2 + (RJ + Lb)\,s + (Rb + k_\mathrm{E}k_\mathrm{M})} \tag{A.2}$$

For the state-space model, the angular position, the angular velocity and the armature current are

selected as state variables, $\mathbf{x} = \begin{bmatrix} \theta & \omega & i \end{bmatrix}^{\mathrm{T}}$. The armature voltage u is the input and the angular position θ is the output. Rearranging Equations (2.1) and (2.2) for the derivatives of the state variables gives the state-space model in Equation (A.3).

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}\,u, \qquad \theta = \mathbf{c}^{\mathrm{T}}\mathbf{x} + d\,u \tag{A.3}$$

with

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\dfrac{b}{J} & \dfrac{k_{\mathrm{M}}}{J} \\ 0 & -\dfrac{k_{\mathrm{E}}}{L} & -\dfrac{R}{L} \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ \dfrac{1}{L} \end{bmatrix}, \qquad \mathbf{c}^{\mathrm{T}} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, \qquad d = 0.$$

Chapter B

Additional Measurement Data

In addition to the step responses, the tracking of a sinusoidal reference with an amplitude of $10°$ and a frequency of $0.8\,\mathrm{Hz}$ is measured for different proportional gains, while K_I and K_D are kept constant. The results are listed in Table B.1.

Table B.1: Control error and control value for the tracking of a sinusoidal reference (amplitude $10°$, frequency $0.8\,\mathrm{Hz}$) with $K_\mathrm{I} = 22.2\,\mathrm{V\,rad^{-1}\,s^{-1}}$ and $K_\mathrm{D} = 4.28\,\mathrm{V\,s\,rad^{-1}}$, evaluated for $t > 2\,\mathrm{s}$

$K_\mathrm{P}/(\mathrm{V\,rad^{-1}})$	RMS of $e/°$	max. $\lvert e\rvert/°$	max. $\lvert u\rvert/\mathrm{V}$
195	0.59	0.86	3.5
300	0.38	0.59	3.6
600	0.19	0.34	4.2
800	0.15	0.37	6.3
1200	0.12	0.42	10.6